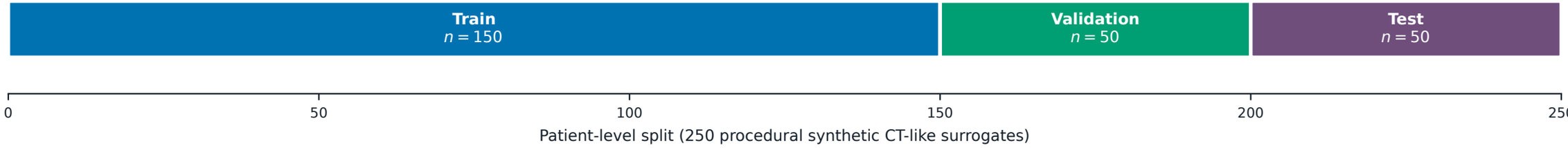
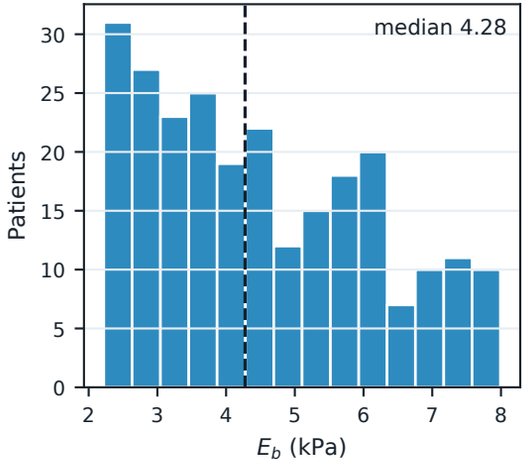
Dataset and observation protocol

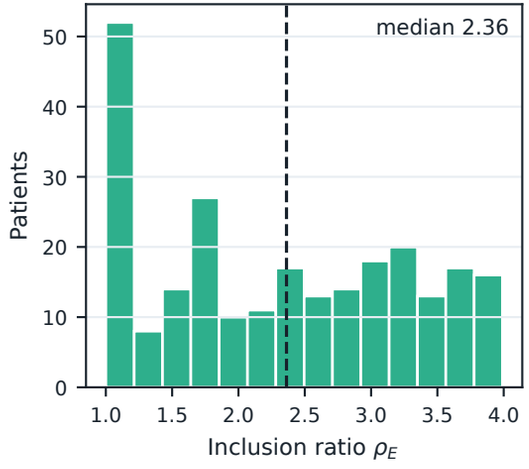
A Development-fixed case-level partition

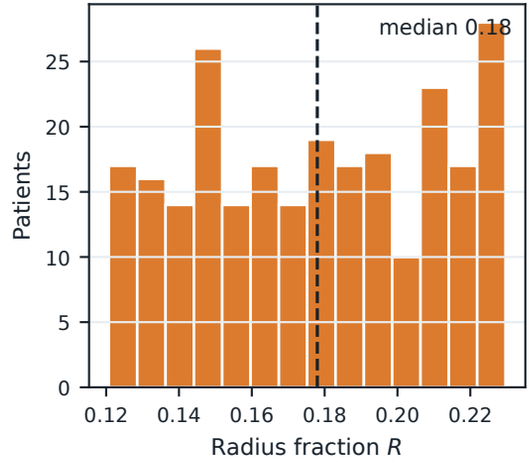
Model selection: train only • interval/prior-scale calibration: validation only • test: after freezing

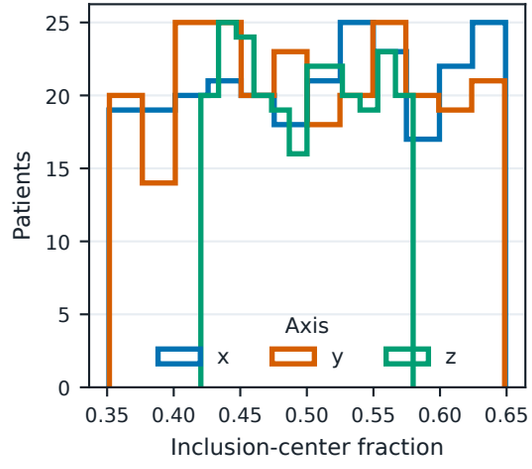


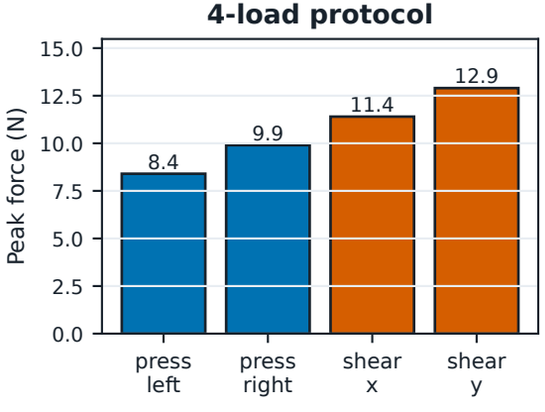
B Generated material/region distributions

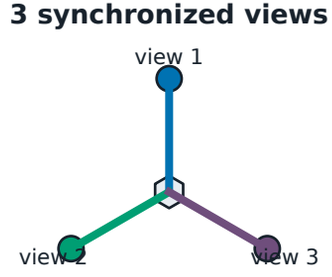


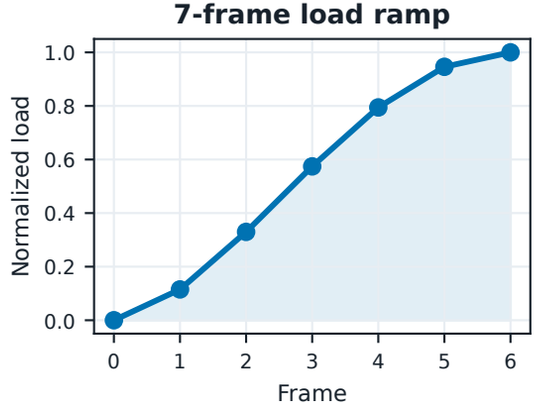












Observation model

Track noise **0.05 px**	Force prior **5%**
Depth change **normalized**	Visibility **binary mask**
Occlusion **confidence**	Camera pose **known**

$$y_{t,v} = \Pi_v(x + u_t) + \varepsilon,\ \varepsilon \sim \mathcal{N}(0, 0.05^2)$$